Answers to these questions

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1. Dividing a program into functions

a. is the key to object-oriented programming.
b. **makes the program easier to conceptualize.**
c. may reduce the size of the program.
d. makes the program run faster.

2. A function name must be followed by parentheses **() .**

3. A function body is delimited curly braces {}.

4. Why is the main() function special?

It is the starting point of every C++ program

5. A C++ instruction that tells the computer to do something is called a Statements.

6**. Write an example of a normal C++ comment and an example of an old-fashioned /***
comment.

Normal C++ comment: **// This is a single-line comment**

Old-fashioned comment: **/* This is a multi-line comment */**

7. An expression

a**. usually evaluates to a numerical value.**
b. indicates the emotional state of the program.
c. always occurs outside a function.
d. may be part of a statement.

8. Specify how many bytes are occupied by the following data types in a 32-bit system:

a. Type int
b. **Type long double**
c. Type float
d. Type long

9. True or false: A variable of type char can hold the value 301.

False

Reason: A char variable only holds 1 byte (8 bits). Its maximum value is 127 (signed) or 255 (unsigned). The number 301 is too large.

10. What kind of program elements are the following?

a. **12:** Integer literal
b. **'a':** Character literal
c. **4.28915:** Floating-point literal (double)
d. **JungleJim:** Identifier (variable or class name)
e. **JungleJim():** Function call (or constructor call)

11. Write statements that display on the screen

a. the character 'x'
b. the name *Jim*
c. the number 509

a. std::cout << 'x';

b. std::cout << "Jim";

c. std::cout << 509;

12. True or false: In an assignment statement, the value on the left of the equal sign is always equal to the value on the right. False

reason The = operator signifies **assignment**, not algebraic equality.

13. Write a statement that displays the variable george in a field 10 characters wide.

std::cout << std::setw(10) << george;

14. What header file must you #include with your source file to use cout and cin?

<iostream>

15. Write a statement that gets a numerical value from the keyboard and places it in the variable temp.

std::cin >> temp;

16. What header file must you #include with your program to use setw?

<iomanip>

17. Two exceptions to the rule that the compiler ignores whitespace are String literals and Preprocessor directives.
18. True or false: It's perfectly all right to use variables of different data types in the same
arithmetic expression. **True**

19. The expression 11%3 evaluates to 2**.**

20. An arithmetic assignment operator combines the effect of what two operators?

assignment (=) with an arithmetic operation (like +, -, *, /, or %)

21. Write a statement that uses an arithmetic assignment operator to increase the value of
the variable temp by 23. Write the same statement without the arithmetic assignment
operator.

temp = temp + 23

22. The increment operator increases the value of a variable by how much?

The increment operator (++) increases the value of a variable by exactly **1**.

23. Assuming var1 starts with the value 20, what will the following code fragment print out?
cout << var1--;
cout << ++var1;

24. In the examples we've seen so far, header files have been used for what purpose?

Header files are used to provide the **declarations** and function prototypes (e.g., cin, cout) required to link prewritten code to your source file.

25. The actual code for library functions is contained in a library file.

Q7. You can convert temperature from degrees Celsius to degrees Fahrenheit by multiplying by 9/5 and adding 32. Write a program that allows the user to enter a floating-point number representing degrees Celsius, and then displays the corresponding degrees Fahrenheit.

```
#include <iostream>
using namespace std;

int main() {
   // Declare floating-point variables
   double celsius;
   double fahrenheit;

   // Prompt user for input
   cout << "Enter temperature in Celsius: ";
   cin >> celsius;

   // Perform conversion using floating-point math (9.0 / 5.0)
   fahrenheit = (celsius * 9.0 / 5.0) + 32.0;

   // Display the result
   cout << "Equivalent temperature in Fahrenheit: " << fahrenheit << endl;

   return 0;
}
```

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1. Assume that you want to generate a table of multiples of any given number. Write a program
that allows the user to enter the number and then generates the table, formatting it
into 10 columns and 20 lines. Interaction with the program should look like this (only the
first three lines are shown):

Enter a number: 7

7	**14**	**21**	**28**	**35**	**42**	**49**	**56**	**63**	**70**
77	**84**	**91**	**98**	**105**	**112**	**119**	**126**	**133**	**140**
147	**154**	**161**	**168**	**175**	**182**	**189**	**196**	**203**	**210**

```
#include <iostream>
```

```
#include <iomanip> // Required for setw() formatting

using namespace std;

int main() {
   int number;

   // Prompt the user for input
   cout << "Enter a number: ";
   cin >> number;

   // The table requires 20 lines (rows) and 10 columns.
   // Total multiples to generate = 20 * 10 = 200.
   for (int i = 1; i <= 200; ++i) {
      // Calculate the current multiple
      int multiple = number * i;

      // Print the multiple with a fixed width of 5 spaces for clean alignment
      cout << setw(5) << multiple;

      // If we have printed 10 columns, move to the next line
      if (i % 10 == 0) {
         cout << "\n";
      }
   }

   return 0;
}
```

2. Write a temperature-conversion program that gives the user the option of converting
Fahrenheit to Celsius or Celsius to Fahrenheit. Then carry out the conversion. Use
floating-point numbers. Interaction with the program might look like this:
Type 1 to convert Fahrenheit to Celsius,
2 to convert Celsius to Fahrenheit: 1
Enter temperature in Fahrenheit: 70
In Celsius that’s 21.111111

```cpp
#include <iostream>

int main() {
    int choice;
    double fahrenheit, celsius;

    // Display options to the user
    std::cout << "Type 1 to convert Fahrenheit to Celsius,\n";
    std::cout << "2 to convert Celsius to Fahrenheit: ";
    std::cin >> choice;

    // Process the user's choice
    if (choice == 1) {
        // Fahrenheit to Celsius conversion
        std::cout << "Enter temperature in Fahrenheit: ";
        std::cin >> fahrenheit;

        celsius = (fahrenheit - 32.0) * 5.0 / 9.0;

        std::cout << "In Celsius that's " << celsius << std::endl;
    }
    else if (choice == 2) {
        // Celsius to Fahrenheit conversion
        std::cout << "Enter temperature in Celsius: ";
        std::cin >> celsius;

        fahrenheit = (celsius * 9.0 / 5.0) + 32.0;

        std::cout << "In Fahrenheit that's " << fahrenheit << std::endl;
    }
    else {
        // Error handling for invalid menu choices
        std::cout << "Invalid choice. Please run the program again and enter 1 or 2." << std::endl;
    }

    return 0;
}
```